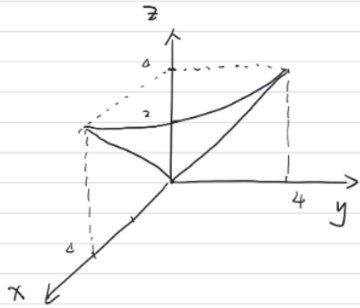


Worked Examples and General Theory

Tuesday, June 17, 2025 7:10 AM

18. $\vec{F} = [xy, yz, zx]$, S the surface of the cone $x^2 + y^2 \leq 4z^2$, $0 \leq z \leq 2$

Q18. $\vec{F} = [xy, yz, zx]$ $S: x^2 + y^2 \leq 4z^2$ $z \in [0, 2]$



$$\iint_S \vec{F} \cdot \vec{n} \, dA = \iiint_T \operatorname{div} \vec{F} \, dV$$

$$\begin{aligned} \operatorname{div} \vec{F} &= \vec{\nabla} \cdot \vec{F} = \left[\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right] \cdot [xy, yz, zx] \\ &= y + z + x \end{aligned}$$

$$\iiint_T (y + z + x) \, dx \, dy \, dz$$

$$\int_{r=0}^{2z} \int_{\theta=0}^{2\pi} \int_{z=0}^2 (r \cos \theta + r \sin \theta + z) r dr d\theta dz$$

$$= \int_{\theta=0}^{2\pi} \int_{z=0}^2 \int_{r=0}^2 r^2 (\cos \theta + \sin \theta) + rz dr d\theta dz$$

$$= \int_{\theta=0}^{2\pi} \int_{z=0}^2 (\cos \theta + \sin \theta) \cdot \frac{1}{3} r^3 \Big|_0^{2z} + z \cdot \frac{1}{2} r^2 \Big|_0^{2z} d\theta dz$$

$$= \int_{\theta=0}^{2\pi} \int_{z=0}^2 (\cos \theta + \sin \theta) \frac{8}{3} z^3 + 2z^3 d\theta dz$$

$$= \int_{\theta=0}^{2\pi} \int_{z=0}^2 \left(\frac{8}{3} (\cos \theta + \sin \theta) + 2 \right) z^3 dz d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(\frac{8}{3} (\cos \theta + \sin \theta) + 2 \right) \frac{1}{4} z^4 \Big|_0^2 d\theta$$

$$= \int_{\theta=0}^{2\pi} 4 \left(\frac{8}{3} (\cos \theta + \sin \theta) + 2 \right) d\theta$$

$$= \int_{\theta=0}^{2\pi} 4 \left(\frac{8}{3} (\cos \theta + \sin \theta) + 2 \right) d\theta$$

$$= \frac{32}{3} \int_{\theta=0}^{2\pi} \cos \theta + \sin \theta d\theta + 8 \int_{\theta=0}^{2\pi} d\theta$$

$$= \frac{32}{3} \cdot \sin \theta - \cos \theta \Big|_0^{2\pi} + 8 \cdot 2\pi$$

$$= \frac{32}{3} [(0-1) - (0-1)] + 16\pi$$

$$= 16\pi$$

23- Spherical and Cylindrical Coordinates.

Table with the del operator in cartesian, cylindrical and spherical coordinates

Operation	Cartesian coordinates (x, y, z)	Cylindrical coordinates (ρ, φ, z)	Spherical coordinates (r, θ, φ), where θ is the polar angle and φ is the azimuthal angle ^a
Vector field $\mathbf{A}$	$A_x \hat{\mathbf{x}} + A_y \hat{\mathbf{y}} + A_z \hat{\mathbf{z}}$	$A_\rho \hat{\boldsymbol{\rho}} + A_\varphi \hat{\boldsymbol{\varphi}} + A_z \hat{\mathbf{z}}$	$A_r \hat{\mathbf{r}} + A_\theta \hat{\boldsymbol{\theta}} + A_\varphi \hat{\boldsymbol{\varphi}}$
Gradient ∇f ^[1]	$\frac{\partial f}{\partial x} \hat{\mathbf{x}} + \frac{\partial f}{\partial y} \hat{\mathbf{y}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}$	$\frac{\partial f}{\partial \rho} \hat{\boldsymbol{\rho}} + \frac{1}{\rho} \frac{\partial f}{\partial \varphi} \hat{\boldsymbol{\varphi}} + \frac{\partial f}{\partial z} \hat{\mathbf{z}}$	$\frac{\partial f}{\partial r} \hat{\mathbf{r}} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \varphi} \hat{\boldsymbol{\varphi}}$
Divergence $\nabla \cdot \mathbf{A}$ ^[1]	$\frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$	$\frac{1}{\rho} \frac{\partial (\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\varphi}{\partial \varphi} + \frac{\partial A_z}{\partial z}$	$\frac{1}{r^2} \frac{\partial (r^2 A_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\varphi}{\partial \varphi}$
Curl $\nabla \times \mathbf{A}$ ^[1]	$\begin{pmatrix} \frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \\ \frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \\ \frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \end{pmatrix} \hat{\mathbf{x}} + \dots$	$\begin{pmatrix} \frac{1}{\rho} \frac{\partial A_z}{\partial \varphi} - \frac{\partial A_\varphi}{\partial z} \\ \frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial \rho} \\ \frac{1}{\rho} \left(\frac{\partial (\rho A_\varphi)}{\partial \rho} - \frac{\partial A_\rho}{\partial \varphi} \right) \end{pmatrix} \hat{\boldsymbol{\rho}} + \dots$	$\begin{pmatrix} \frac{1}{r \sin \theta} \left(\frac{\partial}{\partial \theta} (A_\varphi \sin \theta) - \frac{\partial A_\theta}{\partial \varphi} \right) \\ \frac{1}{r} \left(\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \varphi} - \frac{\partial}{\partial r} (r A_\varphi) \right) \\ \frac{1}{r} \left(\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right) \end{pmatrix} \hat{\mathbf{r}} + \dots$
Laplace operator $\nabla^2 f \equiv \Delta f$ ^[1]	$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$	$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \varphi^2} + \frac{\partial^2 f}{\partial z^2}$	$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \varphi^2}$

2.3.2 General Theory

To simplify formulas, we write Cartesian coordinates $x = x_1, y = x_2, z = x_3$. We denote curvilinear coordinates by q_1, q_2, q_3 . Through each point P there pass three coordinate surfaces $q_1 = \text{const}, q_2 = \text{const}, q_3 = \text{const}$. They intersect along coordinate curves. We assume the three coordinate curves through P to be **orthogonal** (perpendicular to each other). We write coordinate transformations as

$$(1) \quad x_1 = x_1(q_1, q_2, q_3), \quad x_2 = x_2(q_1, q_2, q_3), \quad x_3 = x_3(q_1, q_2, q_3).$$

Corresponding transformations of grad, div, curl, and ∇^2 can all be written by using

$$(2) \quad h_j^2 = \sum_{k=1}^3 \left(\frac{\partial x_k}{\partial q_j} \right)^2.$$

Linear Element ds . In Cartesian coordinates,

$$ds^2 = dx_1^2 + dx_2^2 + dx_3^2$$

For the q -coordinates,

$$(5) \quad ds^2 = h_1^2 dq_1^2 + h_2^2 dq_2^2 + h_3^2 dq_3^2.$$

Gradient. $\text{grad } f = \nabla f = [f_{x_1}, f_{x_2}, f_{x_3}]$ (partial derivatives; Sec. 9.7). In the q -system, with $\mathbf{u}, \mathbf{v}, \mathbf{w}$ denoting unit vectors in the positive directions of the q_1, q_2, q_3 coordinate curves, respectively,

$$(6) \quad \text{grad } f = \nabla f = \frac{1}{h_1} \frac{\partial f}{\partial q_1} \mathbf{u} + \frac{1}{h_2} \frac{\partial f}{\partial q_2} \mathbf{v} + \frac{1}{h_3} \frac{\partial f}{\partial q_3} \mathbf{w}$$

Divergence $\operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = (F_1)_{x_1} + (F_2)_{x_2} + (F_3)_{x_3}$ ($\mathbf{F} = [F_1, F_2, F_3]$, Sec. 9.8);

$$(7) \quad \operatorname{div} \mathbf{F} = \nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} (h_2 h_3 F_1) + \frac{\partial}{\partial q_2} (h_3 h_1 F_2) + \frac{\partial}{\partial q_3} (h_1 h_2 F_3) \right]$$

Laplacian $\nabla^2 f = \nabla \cdot \nabla f = \operatorname{div} (\operatorname{grad} f) = f_{x_1 x_1} + f_{x_2 x_2} + f_{x_3 x_3}$ (Sec. 9.8):

$$(8) \quad \nabla^2 f = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial q_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial f}{\partial q_1} \right) + \frac{\partial}{\partial q_2} \left(\frac{h_3 h_1}{h_2} \frac{\partial f}{\partial q_2} \right) + \frac{\partial}{\partial q_3} \left(\frac{h_1 h_2}{h_3} \frac{\partial f}{\partial q_3} \right) \right]$$

Curl (Sec. 9.9):

$$(9) \quad \operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \mathbf{u} & h_2 \mathbf{v} & h_3 \mathbf{w} \\ \frac{\partial}{\partial q_1} & \frac{\partial}{\partial q_2} & \frac{\partial}{\partial q_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix}.$$

233

Next to Cartesian coordinates, most important are **cylindrical coordinates** $q_1 = r$, $q_2 = \theta$, $q_3 = z$ (Fig. 560a) defined by

$$(3) \quad x_1 = q_1 \cos q_2 = r \cos \theta, \quad x_2 = q_1 \sin q_2 = r \sin \theta, \quad x_3 = q_3 = z$$

and **spherical coordinates** $q_1 = r$, $q_2 = \theta$, $q_3 = \phi$ (Fig. 560b) defined by⁴

$$(4) \quad \begin{aligned} x_1 &= q_1 \cos q_2 \sin q_3 = r \cos \theta \sin \phi, & x_2 &= q_1 \sin q_2 \sin q_3 = r \sin \theta \sin \phi \\ x_3 &= q_1 \cos q_3 = r \cos \phi. \end{aligned}$$

For cylindrical coordinates we have in (9) (as in the previous formulas)

$$h_1 = h_r = 1, \quad h_2 = h_\theta = q_1 = r, \quad h_3 = h_z = 1$$

and for spherical coordinates we have

$$h_1 = h_r = 1, \quad h_2 = h_\theta = q_1 \sin q_3 = r \sin \phi, \quad h_3 = h_\phi = q_1 = r.$$

23.4 Exercise. Use the Lamé coefficients and the given formulae to derive the expressions for ds^2 , $\vec{\nabla} f$, $\vec{\nabla} \cdot \vec{F}$ and $\vec{\nabla} \times \vec{F}$ in both cylindrical and spherical coordinates.